

Chapter 8: Static Electricity

Textbook Exercise Solutions & Answers



1. Choose the correct option and fill in the blanks.

(always repulsion, always attraction, displacement of negative charge, displacement of positive charge, atom, molecule, steel, copper, plastic, inflated balloon, charged object, gold)

(a) There is between like charges.

always repulsion

(b) is responsible for generation of electric charge in an object.

displacement of negative charge

(c) A lightning conductor is made of a strip.

copper

(d) does not get electrically charged easily by rubbing.

Steel

(e) There is when opposite electric charges come near each other.

always attraction

(f) A can be detected with an electroscope.

charged object

2. Explain why it is not safe to go out with an umbrella when there is heavy rain lightning or thunder.

Large amount of charges get built up in the clouds in rainy and stormy weather.

1. Sometimes these charges are transferred to the earth in the form of a bolt of lightning.
2. Wire of umbrella is made up of metal and cloth is made up of synthetic material.
3. If lightning strikes, then umbrella catches it and there are chances of fire and it would be dangerous.
4. So it is not safe to go out with an umbrella when there is heavy rain.

3. Answer in your own words.

(a) How will you protect yourself from lightning?

1. To protect from the lightning, we have to put lightning conductor on high rise buildings,
2. we should not stand under a tree in rainy and stormy weather.
3. We should not go out with an umbrella in heavy rain, lightning or thunder.

(b) How are charges generated?

1. Friction causes a charge to develop, because on rubbing, some negative particles are transferred from one substance to the other and the latter develops a negative charge.
2. The substance from which particles are removed loses negative charge and hence, it develops a positive charge.

(c) In the lightning conductor, what provision is made for spreading the electricity into the ground?

1. The lightning conductor, consists of a long copper strip with one end forked. This end is 'at the highest part of the building.
2. The other end of the strip is connected to a plate of cast iron.
3. A pit is dug in the ground, coal and salt are filled into the pit and the iron plate is placed upright in the pit.
4. There is also a provision for pouring water into it.
5. This helps to spread the electric charge quickly into the ground.

(d) Why do farmers stick an iron staff into the ground while working in the field in rainy conditions?

1. Large amount of charges get built up in the clouds in rainy and stormy weather.
2. Sometimes these charges are transferred to the earth in the form of a bolt of lightning.
3. Due to iron staff, the charges will be conducted to the earth through iron metal and spread quickly into the ground.
4. This way farmers are saved from lightning strike.

(e) Why is lightning not seen every day during the rainy season?

1. When air and clouds rub against each other in the sky the upper part of some clouds becomes positively charged and the lower side becomes negatively charged.
2. When this negative charge at the bottom of the cloud becomes much larger than the charge on the ground, it starts flowing towards the ground in stages, but this doesn't happen every day. So lightning is not seen every day during the rainy season.

4. What are the characteristics of a static electric charge?

Following are the characteristics of a static electric charge.

1. The electric charge generated by friction is called frictional electricity. This charge is produced only at the place of friction, hence it is called static electricity.
2. It remains on the object for a short duration.
3. The charges of static electricity are absorbed in moist air.
4. An electrically charged object attracts another object which has no electric charge.
5. Like charges repel each other.
6. Unlike charges attract each other.
7. Repulsion is the sure test of the existence of a static charge.

5. What is the damage caused by lightning? How will you create awareness to prevent it?

Due to lightning great damage can occur.

1. It may cause fire.
2. Person on whom lightning strikes may die also.
3. It causes more temperature than sun.

Awareness to prevent it:

1. Guide the people to install lightning conductor on tall buildings.
2. Do not stand on open ground, below a tree, or on a high location during the thunder storms.
3. Do not use plugged in electrical appliances.
4. Do not stand near an electric pole, telephone pole or a tower.
5. Stay in pucca house made of brick and cement.
6. Stand on only wood, a plastic sheet or sack of jute.